

Ch. CELL-THE UNIT OF LIFE

Cell Theory:

Formulated by
Schleiden & Schwann
(Botanist) (zoologist)

Modified by

Rudolf Virchow
Omnis-Cellula-e-Cellula

Cell = Fundamental, structural & functional unit of life.

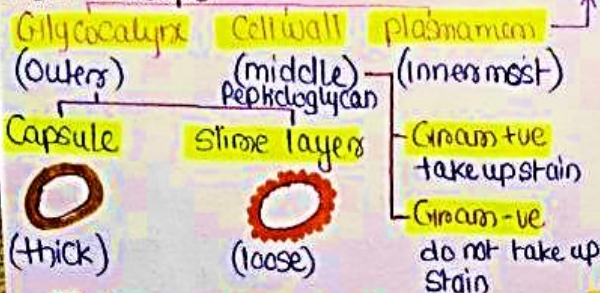
Robert Hooke: discovered dead cells

Antonie van Leeuwenhoek: live cell

"All living cells are made up of cells and its products. all new cells arise from Pre-existing cells."

Prokaryotic Cell:

Cell Envelope:



Chromatophores:

- Contain pigments
- help in photosynthesis

Nucleoid:

- Genetic material
- Circular + ds

Plasmid:

- Extra chromosomal DNA
- Circular + ds.

Flagella:



Pili:



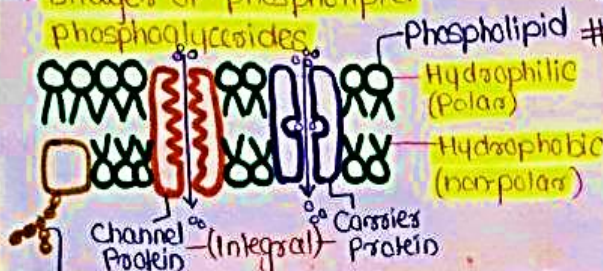
Fimbriae:



Eukaryotic Cell:

Plasma-Membrane:

- Bilayers of phospholipid phosphoglycerides



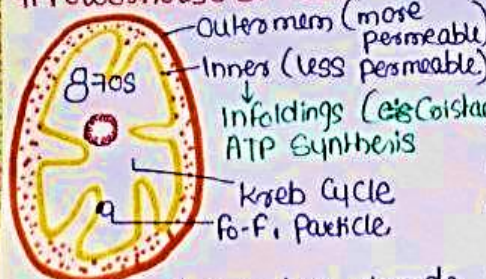
Cholesterol (Peripheral)

Cell wall:

- Fungi = Chitin
- Plants = Cellulose, hemicellulose, pectin
- Algae = Cellulose, Galactans, Mannans

Mitochondria:

Powerhouse of cell



- > no. of mitochondria depends on activity of cell
- > Chloroplast & mitochondria

Semi autonomous

Endomembrane System:

- Organelles work in co-ordinated manner

Endoplasmic Reticulum:

- RER**
 - mem. Continuous with nuclear envelope
 - 80S Ribosome attached
 - Protein Synthesis
- SER**
 - freely present in cytoplasm
 - lack ribosome
 - Lipid Synthesis

Vacuole:

- Single mem Tonoplast
- Plants - large Central
- Contractile - Amoeba
- Excretion & osmoregulation
- Gas - buoyancy
- Green & purple bacteria
- Food vacuole - engulf food; protist

Lysosome:

- Suicidal bags
- pH - acidic (4-5)
- formed by vesicles of Golgi
- Hydrolytic enzyme
- ① Protease
- ② Lipase
- ③ Nuclease
- ④ Carbohydrase

Golgi bodies:

dichytosome → plants

- # Camilla Golgi**
- forming / cis / convex

- protein + lipid
- Glycosylation
- packaging
- phospholipid + glycolipid + glycoprotein



Plastids:

Exclusively present in plants & Euglenoids.

Leucoplast

- > Colorless
- ① **Amyloplast**
Store amylase + amylopectin

② Elaioplast

Store oils

③ Aleuroplast

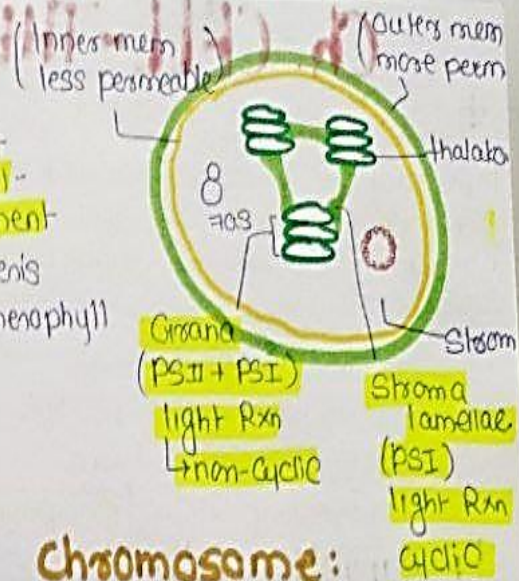
Store protein

Chromoplast

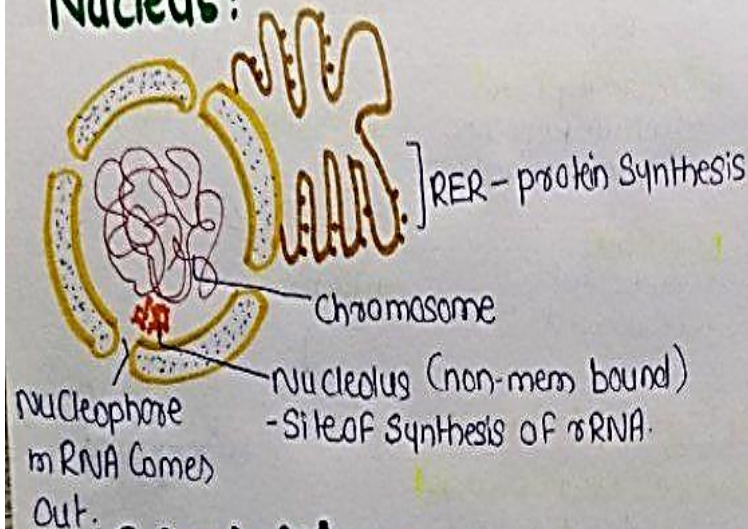
- > Colorless
- > Xanthophyll, Carotenoids

Chloroplast

- > Chlorophyll - green pigment
- > Photosynthesis
- > Found in mesophyll cell



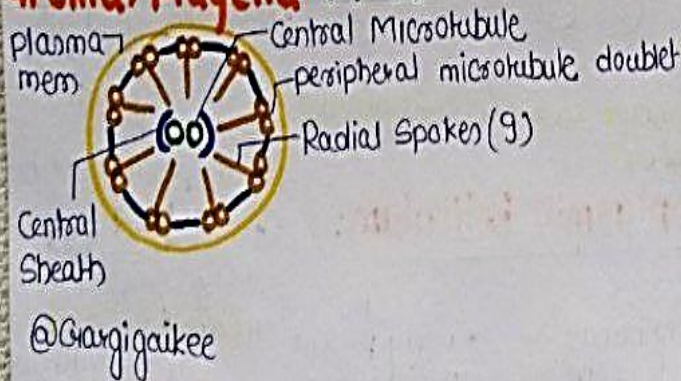
Nucleus:



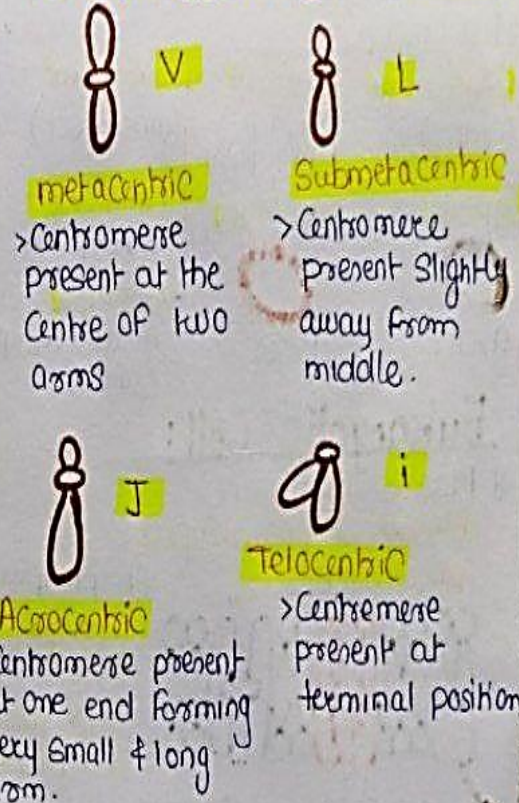
Cytoskeleton:

- Network of proteinaceous filament which provide mechanical support, shape of cell & motility
- Present in Cytoplasm.

Cilia/Flagella (9+2)



Chromosome:



Centrosome & Centriole

